

Dingzhong Zhang

Senior Algorithm Engineer

Foster City, CA

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[Dzzhang96.github.io](https://github.com/Dzzhang96)

Profile

Senior Algorithm Engineer with expertise in medical image processing, surgical navigation and robotics. Specializing in software enhancement via deep learning models and algorithm development. Capable of the design, implementation, and testing of biomedical devices and robotic systems.

Work Experience

Senior Algorithm Engineer

June 2026 – Current

Johnson & Johnson MedTech, MONARCH™ Robotic-Assisted Bronchoscopy | Algorithms & AI Team Santa Clara, CA

- Developed an automated CBCT nodule segmentation framework to enhance biopsy yields by compensating for CT-to-body divergence.
- Post-trained NVGenerate to synthesize abdominal CT images with controllable anatomy size and paired segmentations mask of kidney stones.

Algorithm Engineer

Jan 2024 – May 2026

Johnson & Johnson MedTech, MONARCH™ Robotic-Assisted Bronchoscopy | Algorithms & AI Team Santa Clara, CA

- Development of an automated scope pose estimation workflow for intra-operative navigation updates. [\[US Patent\]](#)
 - a. Reduced the procedure time by over 90% with sub-millimeter accuracy in MONARCH™ Intra-Operative Planning Software.
 - b. Designed a two-stage model inferencing strategy for fast and accurate scope tip segmentation followed by a robust context-aware pose estimation algorithm.
 - c. Evaluated performance on real-world patient CBCT scans from multiple manufacturers, with emphasis on robustness in corner-case scenarios.
 - d. Optimized algorithm performance and inference speed using TensorRT C++ and OpenMP parallel computing.
- Model training for kidney stone segmentation and proposed metrics for stone burden estimation. [\[US Patent\]](#)

R&D Leadership Development Program Engineer

Aug 2022 – Dec 2023

Johnson & Johnson MedTech, MONARCH™ Robotic-Assisted Bronchoscopy | Algorithms & AI Team Redwood City, CA

- End-to-end development of the “one-click” lung nodule segmentation in MONARCH™ Pre-Operative Planning Software. [\[US Patent\]](#) [\[CHEST 2026\]](#)
 - a. Benchmarked and fine-tuned SOTA models for lung nodule segmentation on Auris clinical datasets.
 - b. Accelerated model inference by integrating TensorRT-based C++ deployment into Pre-Op Planning software.
 - c. Built visualization and image-processing framework using VTK/ITK, and implemented corner case solutions.
 - d. Dockerized the pipeline and proposed metrics for precise nodule size measurement in Physician Portal.
- Gateway data mining to validate the hypothesis of a near-infrared fluorescence visualization system using Query for statistical analysis of case metrics. [\[CHEST 2023\]](#)

Algorithm Development Intern

June 2021 – Aug 2021

Medtronic, MTC – Visualization | Robotics Team

- Distortion correction and calibration of intra-operative spine X-ray images using DLT algorithm for Mazor™ system.
- 2D-3D image registration between X-ray and DRR images for C-Arm image-guided surgical navigation.
- Implemented an iterative UNet for spine segmentation, including partially visible structures at image boundaries.

Research Assistant

Dec 2018 – July 2019

Orthotek Laboratory, Prof. Zikai Hua

Shanghai, China

- Established a universal test platform for simulating various kinematic and load scenarios in prosthetic knee joints.
 - a. Performed kinematic and load simulations using an ABB IRB 6700 robotic arm for prosthetic knee evaluation.
 - b. Improved prosthesis control strategies by analyzing load conditions using a multi-dimensional force sensor, a Beckhoff embedded PC, and custom software developed on TwinCAT.
 - c. Developed a dynamic optical motion measurement system to capture prosthetic knee kinematics during gait using surface-mounted markers.

Publications - [Google Scholar](#)

- **Dingzhong Zhang**, Lauren Friend, Babak Matinfar, Siobhan Carroll, Samaan Rafeq, Matthew Krever*. (2023). Achievable Distance to Nodule Measurement in Real-world Robotic-assisted Bronchoscopy Data. CHEST 2023. [\[Poster Presentation\]](#)
- Jiangchang Xu, **Dingzhong Zhang**, Chunliang Wang, Huifang Zhou, Yinwei Li, and Xiaojun Chen*. (2023). Automatic Segmentation of Orbital Wall from CT Images Via a Thin Wall Region Supervision-Based Multi-scale Feature Search Network. International Journal of Computer Assisted Radiology and Surgery [\[Link\]](#); 2023 CARS [\[Poster Presentation\]](#)
- Yuwei Gu, **Dingzhong Zhang**, Baoxin Tao, Feng Wang, Xiaojun Chen*, Yiqun Wu*. (2023). A novel technique to quantify radiographic bone-to-implant contact of zygomatic implants based on three-dimensional image registration and segmentation. Dentomaxillofacial Radiology. [\[Link\]](#)
- **Dingzhong Zhang**, Ahmed Aoude, Mark Driscoll*. (2022). Development and Model Form Assessment of an Automatic Subject-specific Vertebra Reconstruction Method. Computers in Biology and Medicine. [\[Link\]](#)
- Jiangchang Xu[#], Jiannan Liu[#], **Dingzhong Zhang**, Zijie Zhou, Chenping Zhang, Xiaojun Chen*. (2021). A 3D Segmentation Network of Mandible from CT Scan with Combination of Multiple Convolutional Modules and Edge Supervision in Mandibular Reconstruction. Computers in Biology and Medicine. [\[Link\]](#)
- Haitao Li[#], Jiangchang Xu[#], **Dingzhong Zhang**, Yaohua He, Xiaojun Chen*. (2022). An Automatic Surgical Planning Combining Bone Density Assessment and Path Integral in Cone Area for Reverse Shoulder Arthroplasty. International Journal of Computer Assisted Radiology and Surgery. [\[Link\]](#)
- Jiangchang Xu[#], Jiannan Liu[#], **Dingzhong Zhang**, Zijie Zhou, Xiaoyi Jiang, Chenping Zhang, Xiaojun Chen*. (2021). Automatic Mandible Segmentation from CT Image Using 3D Fully Convolutional Neural Network Based on DenseASPP and Attention Gates. International Journal of Computer Assisted Radiology and Surgery. [\[Link\]](#)
- Afaque R. Memon, Dongyuan Li, Junlei Hu, Enpeng Wang, **Dingzhong Zhang**, Xiaojun Chen*. (2021). The development of computer-aided patient-specific template design software for 3D printing in cranio-maxillofacial surgery. The International Journal of Medical Robotics and Computer Assisted Surgery. [\[Link\]](#)

Education

McGill University	Sept 2020 – Oct 2022
• Master of Science – Mechanical Engineering (GPA: 4.00 / 4.00)	
Shanghai Jiao Tong University	Sept 2019 – Sept 2021
• Master of Science – Mechanical Engineering (GPA: 3.64 / 4.00)	
Shanghai University	Sept 2014 – June 2019
• Bachelor of Engineering – Mechanical Engineering (GPA: 3.72 / 4.00 RANK: 1/277)	

Academic Research

Augmented Reality-based Spine Surgical Navigation System via Polaris & HoloLens	Oct 2020 – Feb 2022
<i>McGill University</i>	Montréal, Canada
• Designed an AR spine surgical navigation system using Polaris and HoloLens for enhanced Intra-Op visualization.	
• Proposed an efficient and accurate workflow for surgical instruments calibration and registration.	
• Trained Ki-UNet for vertebra segmentation and validated accuracy using FaroArm coordinate measurements.	
[CIBM 2022]	
Orbital Rim Registration	Jan 2020 – June 2020
<i>Shanghai Jiao Tong University, Shanghai Ninth People's Hospital, Prof. Xiaojun Chen</i>	Shanghai, China
• Developed a C++ software application that enables physicians to register anatomical landmarks along the orbital rim for ocular surgical procedures using Qt and VTK.	
• Implemented the ICP algorithm for surface registration between patient anatomy and reconstructed model from CT imaging data.	
Bone Graft Segmentation in Maxillary Sinus	Sept 2019 – Nov 2019
<i>Shanghai Jiao Tong University, Prof. Xiaojun Chen</i>	Shanghai, China
• Reduced computation time of a medical image computation software by 80% through integration of DL models.	
• Compiled the TensorFlow C++ library and translated the segmentation pipelines from Python to C++ for deployment and performance optimization.	
• Developed semi-automatic labelling algorithms to segment the bone graft from maxillary sinus for training.	
Design of a Medical Rehabilitation Robotic Arm	Jan 2018 – Jun 2018

- Designed and prototyped a 4-DOF rehabilitation robotic arm capable of predicting patient arm movements and adaptively adjusting assistance levels during therapeutic exercises.
- Developed the complete robotic system featuring multi-sensor feedback, servo control, and a Windows Forms-based control interface.
- Conducted FEA to optimize the mechanical structure for enhanced stiffness and load-bearing performance.

Awards & Honors

- Grad Excellence Award - McGill University 2021
- Academic Excellence Scholarship - Shanghai Jiao Tong University 2019 - 2020
- Graduated with Honors of Shanghai 2019
- Academic Scholarship - Shanghai University 2015 - 2018
- First Prize - Chinese Service Robot Competition 2018
- First Prize - China Aerial Robotics Competition 2018

Languages

- English (Fluent)
- French (B1)
- Chinese (Fluent)

REFERENCES AVAILABLE UPON REQUEST